Cyclic Analytic Functions, Reciprocal Families, and Zero Geometry

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Abstract

Fix a positive integer N. Cyclic projection separates an analytic function into coefficient classes modulo N. Applied to the exponential, it gives a basis for $y^{(N)} = \lambda^N y$. A signed basis for $y^{(N)} = -y$ is realized inside the finite algebra $\mathbb{C}[\xi]/(\xi^N + 1)$ and produces the entire denominator

$$\mathsf{D}_N(z) = \sum_{n\geq 0} (-1)^n \frac{z^{Nn}}{(Nn)!}.$$

Its reciprocal coefficients are polynomial in the reciprocal order. We derive Bell polynomial, Newton recurrence, and determinant formulas. Normalized sections give Appell systems with operator and two parameter convolution laws. Writing $\mathsf{D}_N(z) = \Delta_N(z^N)$ reduces the zero problem to order $1/N$. We determine the indicator polygon, locate every sufficiently large zero, obtain its first exponentially small displacement, prove eventual simplicity, and derive the exact zero count. The reduced denominator has a genus zero product and a trace class determinant model. Its zero power sums recover the reciprocal coefficients and logarithmic derivative. The symmetry grouped zero zeta continues meromorphically by comparison with the Hurwitz zeta function.

1. Introduction

For a power series

$$F(z) = \sum_{m\geq 0} a_m z^m$$

and a positive integer N, the coefficients may be separated according to the residue of m modulo N. Root of unity averaging gives exact projection operators for this separation. This elementary construction is usually called a multisection or a root filter. It appears throughout the classical theory of generating functions

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and special functions. Background on multisections, hypergeometric functions, and related coefficient systems may be found in [Rai60], [AAR99], and [Com74].

This article follows the consequences of applying the cyclic projections to the exponential function. The projected exponentials form a finite differential system. After one sign is inserted around the derivative cycle, they form a basis for

$$y^{(N)} = -y.$$

The section with initial value one is

$$\mathsf{D}_N(z) = \sum_{n\geq 0} (-1)^n \frac{z^{Nn}}{(Nn)!}.$$

Its reciprocal powers produce polynomial coefficient families. Its logarithmic derivative produces a higher Riccati equation. The same reciprocal construction applied to every exponential section gives normalized congruential functions and ordinary Appell polynomial sequences. Appell and binomial type sequences are discussed in [Rom84] and [Com74].

The second half of the paper concerns the zeros of D_N. The finite exponential sum

$$\mathsf{D}_N(z) = \frac{1}{N} \sum_{\eta^N=-1} e^{\eta z}$$

has a regular polygon of exponents. The general relation between an exponent polygon and the zero strips of an exponential sum goes back to Pólya and Langer. Later developments include the work of Moreno and the treatment in Levin's monograph [Pól20], [Lan31], [Mor73], and [Lev80]. In the present family the regular polygon and conjugation symmetry permit an explicit specialization of this theory. Two adjacent exponentials dominate on each of N rays. Their cancellation gives one zero string on every ray. We prove that these strings contain every sufficiently large zero, that the displacement from the two term model is exponentially small, and that every zero in the strings is simple. This gives an exact linear zero counting law.

The rotational symmetry is more useful after the change of variable $w = z^N$. There is an entire function

$$\Delta_N(w) = \sum_{n\geq 0} (-1)^n \frac{w^n}{(Nn)!}$$

with

$$\mathsf{D}_N(z) = \Delta_N(z^N).$$

For $N \geq 2$, the order of Δ_N is $1/N$. Its Hadamard product therefore has genus zero. The logarithm of this product identifies the transfer tangent coefficients with power sums of the reduced zeros. The large zero formula also compares the reduced zero zeta with a Hurwitz zeta function. Standard material on

entire functions, canonical products, and exponential sums appears in [Boa54], [Lev80], and [Tit39]. We use singularity analysis in the form described in [FS09] and the classical continuation of the Hurwitz zeta function from [Apo76].

The article does not assign historical priority to every named coefficient family. Several low order cases are classical, including the exponential, cosine, secant, tangent, Euler, Bernoulli type, and Appell cases. The point is to place the full cyclic family in one analytic structure and to prove the relations among its differential equations, reciprocal powers, zeros, and coefficient systems.

2. **A map of the construction**

Two variables appear throughout the paper. The variable z carries the differential equation and the rotational symmetry. The reduced variable

$$w = z^N$$

records one representative from each rotational orbit. Confusing these variables hides the main simplification. The denominator D_N has order one as a function of z, while Δ_N has order $1/N$ as a function of w.

The analytic objects are introduced in the following order. First,

$$\mathsf{E}_{N,j}(z) = \sum_{n\geq 0} \frac{z^{Nn+j}}{(Nn+j)!}$$

is the residue j part of the exponential. Second,

$$\mathsf{D}_{N,j}(z) = \sum_{n\geq 0} (-1)^n \frac{z^{Nn+j}}{(Nn+j)!}$$

is the signed basis for $y^{(N)} = -y$. The section $\mathsf{D}_N = \mathsf{D}_{N,0}$ is the denominator used in the reciprocal family

$$\mathsf{S}_N^{(r)} = \mathsf{D}_N^{-r}.$$

Its logarithmic derivative is T_N.

A different reciprocal family comes from normalizing each exponential section at the origin. The function

$$\mathsf{C}_{N,j}(z) = \frac{z^j/j!}{\mathsf{E}_{N,j}(z)}$$

is a unit series in z^N. Its powers, after multiplication by e^{xz}, give Appell polynomials. The theta family couples a scaled argument with a reciprocal order.

The zero theory begins only after these algebraic relations have been established. The factorization

$$\mathsf{D}_N(z) = \Delta_N(z^N)$$

turns every orbit of N zeros in the z plane into one zero in the w plane. The large zeros in the z plane are located by two dominant exponentials. The zeros in the w plane then give a genus zero product. The logarithm of that product returns to the coefficient theory through

$$\text{zero power sums} = \text{logarithmic coefficients}.$$

This equality is the bridge between the first and second halves of the paper.

All powers with a complex parameter are first defined as formal series at the origin. An analytic interpretation requires a simply connected zero free region and one fixed logarithm. Integer reciprocal powers are globally meromorphic and do not require a branch choice.

The main statements fall into five groups. The finite algebra model in section 6 explains the derivative cycles and addition laws through one exponential identity. The coefficient theory in section 7 gives two complementary recurrences, one from the original denominator and one from its logarithm. The Appell theory in section 9 supplies both operator formulas and a convolution law in the space and order parameters. The zero analysis in section 11 gives the indicator polygon, the exact large zero rays, and a first correction term. The product theory in section 12 identifies the logarithmic coefficients with traces of powers of a trace class diagonal operator.

The article keeps formal and analytic statements separate. A formal identity needs only the unit constant term of the relevant series. A holomorphic complex power also needs a zero free domain and a chosen logarithm. A meromorphic coefficient asymptotic is stated only for positive integral reciprocal order unless an additional branch domain is specified.

3. **Cyclic projection of analytic functions**

Fix $N \geq 1$ and write

$$\omega = e^{2\pi i/N}.$$

Let Ω be a disk centered at the origin, with the whole plane allowed, and assume that Ω is invariant under multiplication by ω. For a holomorphic function F on Ω, define

$$(\mathcal{B}_{N,j}F)(z) = \frac{1}{N}\sum_{\ell=0}^{N-1} \omega^{-j\ell}F(\omega^{\ell}z), \qquad 0 \leq j < N.$$

The same formula is valid for formal power series.

Theorem 3.1 (Root filter). *If*

$$F(z) = \sum_{m\geq 0} a_m z^m,$$

then

$$\mathcal{B}_{N,j}F(z) = \sum_{m \equiv j \bmod N} a_m z^m.$$

The operators satisfy

$$\mathcal{B}_{N,i}\mathcal{B}_{N,j} = \delta_{ij}\mathcal{B}_{N,j}$$

and

$$\sum_{j=0}^{N-1} \mathcal{B}_{N,j} = I.$$

Proof. Substitution of the power series gives

$$\mathcal{B}_{N,j}F(z) = \sum_{m \geq 0} a_m z^m \left(\frac{1}{N} \sum_{\ell=0}^{N-1} \omega^{(m-j)\ell} \right).$$

The inner average is one when $m \equiv j \bmod N$ and zero otherwise. The operator identities follow from the disjoint residue classes. □

PROPOSITION 3.2 (Covariance characterization). *A holomorphic function F belongs to the range of $\mathcal{B}_{N,j}$ if and only if*

$$F(\omega z) = \omega^j F(z).$$

Proof. Every monomial selected by $\mathcal{B}_{N,j}$ has the stated covariance. Conversely, write $F = \sum_k \mathcal{B}_{N,k}F$. Replacing z by ωz gives

$$\sum_k \omega^k \mathcal{B}_{N,k}F = \omega^j \sum_k \mathcal{B}_{N,k}F.$$

Application of $\mathcal{B}_{N,k}$ shows that $\mathcal{B}_{N,k}F = 0$ for $k \neq j$. □

Multiplication is not diagonal in these coordinates. The packet labels add modulo N.

THEOREM 3.3 (Cyclic product law). *If F and G are holomorphic on the same rotation invariant domain, then*

$$\mathcal{B}_{N,k}(FG) = \sum_{i+j \equiv k \bmod N} (\mathcal{B}_{N,i}F)(\mathcal{B}_{N,j}G).$$

Proof. A product of a monomial of degree congruent to i and a monomial of degree congruent to j has degree congruent to $i+j$. Summing the corresponding products gives the formula. □

Remark 3.4. Every identity in this section holds formally. In the holomorphic setting, the finite projector sums preserve local uniform convergence. No choice of branch is involved.

4. **The transfer exponential basis**

Definition 4.1. For $\lambda \in \mathbb{C}$ and $0 \leq j < N$, define

$$\mathsf{E}_{N,j}(\lambda z) = \mathcal{B}_{N,j}(e^{\lambda z}) = \sum_{n \geq 0} \frac{(\lambda z)^{Nn+j}}{(Nn+j)!}.$$

THEOREM 4.2 (Differential cycle and initial data). *For* $1 \leq j < N$,

$$\frac{d}{dz}\mathsf{E}_{N,j}(\lambda z) = \lambda \mathsf{E}_{N,j-1}(\lambda z),$$

while

$$\frac{d}{dz}\mathsf{E}_{N,0}(\lambda z) = \lambda \mathsf{E}_{N,N-1}(\lambda z).$$

Thus

$$\frac{d^N}{dz^N}\mathsf{E}_{N,j}(\lambda z) = \lambda^N \mathsf{E}_{N,j}(\lambda z).$$

For $0 \leq k < N$,

$$\left.\frac{d^k}{dz^k}\mathsf{E}_{N,j}(\lambda z)\right|_{z=0} = \delta_{kj}\lambda^j.$$

If $\lambda \neq 0$, *the* N *functions* $\mathsf{E}_{N,j}(\lambda z)$ *form a basis of the solution space of* $y^{(N)} = \lambda^N y$.

Proof. Termwise differentiation gives the cyclic derivative formulas. One full cycle contributes λ^N. The first nonzero term of $\mathsf{E}_{N,j}(\lambda z)$ is $(\lambda z)^j/j!$, which gives the initial data. The initial value vectors are linearly independent when $\lambda \neq 0$. The solution space of an order N constant coefficient equation has dimension N, so these functions form a basis. □

COROLLARY 4.3 (Wronskian of the transfer basis). *For* $\lambda \neq 0$, *the Wronskian of the ordered basis*

$$\mathsf{E}_{N,0}(\lambda z), \ldots, \mathsf{E}_{N,N-1}(\lambda z)$$

is the constant

$$\lambda^{N(N-1)/2}.$$

Proof. At the origin, the derivative matrix is diagonal with entries

$$1, \lambda, \ldots, \lambda^{N-1}.$$

Abel's identity shows that the Wronskian is constant because the coefficient of $y^{(N-1)}$ in the differential equation is zero. □

COROLLARY 4.4 (Addition law). *For* $0 \leq k < N$,

$$\mathsf{E}_{N,k}(\lambda(z+w)) = \sum_{i+j \equiv k \bmod N} \mathsf{E}_{N,i}(\lambda z)\mathsf{E}_{N,j}(\lambda w).$$

Proof. Apply the diagonal cyclic projector to

$$e^{\lambda(z+w)} = e^{\lambda z} e^{\lambda w}.$$

The cyclic product law gives the right hand side. □

The basis also has a scalar hypergeometric form.

THEOREM 4.5 (Hypergeometric form). *Let*

$$A_{N,j} = \left\{ \frac{j+m}{N} : 1 \le m \le N \text{ and } j+m \ne N \right\}.$$

Then

$$\mathsf{E}_{N,j}(\lambda z) = \frac{(\lambda z)^j}{j!} {}_0F_{N-1}\left(\begin{matrix} - \\ A_{N,j} \end{matrix} , \frac{(\lambda z)^N}{N^N} \right).$$

Proof. The multiplication formula for the gamma function gives

$$(Nn+j)! = j! N^{Nn} n! \prod_{a \in A_{N,j}} (a)_n.$$

Insertion into the defining series gives the stated hypergeometric expansion. □

Example 4.6. For $N = 1$, the only section is $e^{\lambda z}$. For $N = 2$,

$$\mathsf{E}_{2,0}(z) = \cosh z, \qquad \mathsf{E}_{2,1}(z) = \sinh z.$$

These are classical instances of the same cyclic basis.

5. **The signed system and the universal denominator**

Definition 5.1. For $0 \le j < N$, define

$$\mathsf{D}_{N,j}(z) = \sum_{n \ge 0} (-1)^n \frac{z^{Nn+j}}{(Nn+j)!}.$$

The universal denominator is

$$\mathsf{D}_N(z) = \mathsf{D}_{N,0}(z).$$

Fix any $\rho \in \mathbb{C}$ with $\rho^N = -1$.

PROPOSITION 5.2 (Relation with the transfer exponential). *For every* j,

$$\mathsf{D}_{N,j}(z) = \rho^{-j} \mathsf{E}_{N,j}(\rho z).$$

Proof. The coefficient of z^{Nn+j} on the right is

$$\rho^{-j} \rho^{Nn+j} \frac{1}{(Nn+j)!} = (-1)^n \frac{1}{(Nn+j)!}.$$

□

THEOREM 5.3 (Signed derivative cycle). *For* $1 \le j < N$,

$$\mathsf{D}'_{N,j} = \mathsf{D}_{N,j-1},$$

and

$$\mathsf{D}'_{N,0} = -\mathsf{D}_{N,N-1}.$$

Every signed section satisfies

$$\mathsf{D}^{(N)}_{N,j} = -\mathsf{D}_{N,j}.$$

The N *signed sections form the basis of* $y^{(N)} = -y$ *determined by*

$$\mathsf{D}^{(k)}_{N,j}(0) = \delta_{jk}$$

for $0 \le j, k < N$.

Proof. The derivative formulas follow directly from the series. The first nonzero term of $\mathsf{D}_{N,j}$ is $z^j/j!$, so the initial value matrix is the identity. The basis statement follows from the dimension of the solution space. □

COROLLARY 5.4 (Wronskian of the signed basis). *The Wronskian of*

$$\mathsf{D}_{N,0}(z), \dots, \mathsf{D}_{N,N-1}(z)$$

is one.

Proof. The derivative matrix at the origin is the identity. Abel's identity shows that the Wronskian is constant because the coefficient of $y^{(N-1)}$ in the differential equation is zero. □

PROPOSITION 5.5 (Twisted addition law). *For* $0 \le k < N$,

$$\mathsf{D}_{N,k}(z+w) = \sum_{\substack{0 \le i,j < N \\ i+j \equiv k \bmod N}} (-1)^{\lfloor (i+j)/N \rfloor} \mathsf{D}_{N,i}(z)\mathsf{D}_{N,j}(w).$$

Proof. Use proposition 5.2 and corollary 4.4. The factor introduced by a pair (i, j) is

$$\rho^{i+j-k}.$$

Since $0 \le i+j \le 2N-2$ and $i+j \equiv k \bmod N$, this factor is one when $i+j = k$ and -1 when $i+j = k+N$. □

THEOREM 5.6 (Finite exponential representation). *Let*

$$\mathcal{R}_N = \{\eta \in \mathbb{C} : \eta^N = -1\}.$$

Then

$$\mathsf{D}_N(z) = \frac{1}{N} \sum_{\eta \in \mathcal{R}_N} e^{\eta z}.$$

More generally,

$$\mathsf{D}_{N,j}(z) = \frac{\rho^{-j}}{N} \sum_{\ell=0}^{N-1} \omega^{-j\ell} e^{\rho\omega^{\ell} z}.$$

Proof. The second formula is the root filter expression for $\rho^{-j}\mathsf{E}_{N,j}(\rho z)$. The first is the case $j = 0$. □

Definition 5.7. Define the reduced denominator

$$\Delta_N(w) = \sum_{n\geq 0} (-1)^n \frac{w^n}{(Nn)!}.$$

Then

$$\mathsf{D}_N(z) = \Delta_N(z^N).$$

Corollary 5.8 (Symmetries). *The denominator satisfies*

$$\mathsf{D}_N(\omega z) = \mathsf{D}_N(z)$$

and

$$\mathsf{D}_N(\overline{z}) = \overline{\mathsf{D}_N(z)}.$$

Its nonzero zeros occur in orbits under multiplication by the Nth roots of unity and under complex conjugation.

Proof. The first identity follows from the powers z^{Nn}. The second follows from the real coefficients. Both symmetries preserve the zero set and multiplicity. □

Example 5.9. The first denominators are

$$\mathsf{D}_1(z) = e^{-z}, \qquad \mathsf{D}_2(z) = \cos z.$$

For $N = 3$,

$$\mathsf{D}_3(z) = \frac{1}{3}\left(e^{-z} + 2e^{z/2}\cos\left(\frac{\sqrt{3}z}{2}\right)\right).$$

For $N = 4$,

$$\mathsf{D}_4(z) = \cos\left(\frac{z}{\sqrt{2}}\right)\cosh\left(\frac{z}{\sqrt{2}}\right).$$

6. **A finite algebra and matrix model**

The derivative and addition laws can be read from one finite algebra identity. This formulation also identifies the exponent polygon as the spectrum of a signed cyclic shift.

Definition 6.1. Let

$$\mathfrak{A}_N^+ = \mathbb{C}[\xi]/(\xi^N - 1)$$

and

$$\mathfrak{A}_N^- = \mathbb{C}[\xi]/(\xi^N + 1).$$

Every element has a unique representative of degree below N.

THEOREM 6.2 (Finite algebra exponential identities). *In $\mathfrak{A}_N^+$ one has*

$$e^{\lambda z \xi} = \sum_{j=0}^{N-1} \mathsf{E}_{N,j}(\lambda z)\xi^j.$$

In $\mathfrak{A}_N^-$ one has

$$e^{z\xi} = \sum_{j=0}^{N-1} \mathsf{D}_{N,j}(z)\xi^j.$$

Differentiation and multiplication of these identities recover the cyclic derivative systems and the two addition laws.

Proof. Expand the exponential and write every exponent uniquely as $Nn+j$. In $\mathfrak{A}_N^+$, the relation $\xi^{Nn+j} = \xi^j$ gives the transfer sections. In $\mathfrak{A}_N^-$, the relation $\xi^{Nn+j} = (-1)^n \xi^j$ gives the signed sections.

Differentiation multiplies by $\lambda\xi$ in the first algebra and by ξ in the second. Reduction of the product by the defining relation gives the derivative cycles. The identity

$$e^{(z+w)\xi} = e^{z\xi} e^{w\xi}$$

gives the addition laws after comparison of the coefficients of $1, \xi, \ldots, \xi^{N-1}$. □

Definition 6.3. Let J_N be the signed cyclic shift on the standard basis $e_0, \ldots, e_{N-1}$ of $\mathbb{C}^N$, defined by

$$J_N e_j = e_{j+1} \quad (0 \le j < N-1), \qquad J_N e_{N-1} = -e_0.$$

Then $J_N^N = -I$.

THEOREM 6.4 (Companion matrix realization). *Put*

$$\mathbf{D}_N(z) = \begin{pmatrix} \mathsf{D}_{N,0}(z) \\ \mathsf{D}_{N,1}(z) \\ \vdots \\ \mathsf{D}_{N,N-1}(z) \end{pmatrix}.$$

Then

$$\mathbf{D}_N(z) = e^{zJ_N} e_0$$

and

$$\mathsf{D}_N(z) = \frac{1}{N} \operatorname{tr}(e^{zJ_N}).$$

The eigenvalues of J_N are the N roots of -1, each with multiplicity one.

Proof. The vector $e^{zJ_N}e_0$ has derivative $J_N e^{zJ_N}e_0$ and initial value e_0. Its components therefore satisfy the same signed derivative cycle and initial data as $\mathbf{D}_N$. Uniqueness for the linear system gives the first identity.

The polynomial $x^N + 1$ has distinct roots, so J_N is diagonalizable with spectrum $\{\eta : \eta^N = -1\}$. Hence

$$\operatorname{tr}(e^{zJ_N}) = \sum_{\eta^N=-1} e^{\eta z} = N\mathsf{D}_N(z)$$

by theorem 5.6. □

Corollary 6.5 (Entries of the fundamental matrix). *For $0 \le p, q < N$, let d_{pq} be the member of $\{0, \dots, N-1\}$ congruent to $p - q$ modulo N. Put*

$$\varepsilon_{pq} = \begin{cases} 1, & p \ge q, \\ -1, & p < q. \end{cases}$$

Then

$$\left(e^{zJ_N}\right)_{pq} = \varepsilon_{pq}\mathsf{D}_{N,d_{pq}}(z).$$

Thus every entry of the fundamental matrix is one of the signed sections, with the sign recording whether the cyclic path from q to p crosses the signed edge.

Proof. For every nonnegative integer n,

$$J_N^{Nn+d_{pq}} e_q = (-1)^n \varepsilon_{pq} e_p.$$

Only powers whose exponent is congruent to d_{pq} modulo N contribute to the (p, q) entry of the exponential. Therefore

$$\left(e^{zJ_N}\right)_{pq} = \varepsilon_{pq} \sum_{n\ge 0} (-1)^n \frac{z^{Nn+d_{pq}}}{(Nn + d_{pq})!}.$$

This is the stated formula. □

Corollary 6.6 (A group law for the fundamental matrix). *Assume $N \ge 2$. The matrix*

$$U_N(z) = e^{zJ_N}$$

satisfies

$$U_N(z+w) = U_N(z)U_N(w), \qquad U_N(z)^{-1} = U_N(-z), \qquad \det U_N(z) = 1.$$

Its first column is $\mathbf{D}_N(z)$, and all of its entries are given by corollary 6.5.

Proof. The first two identities are the exponential group law. Since $\operatorname{tr} J_N = 0$,

$$\det(e^{zJ_N}) = e^{z \operatorname{tr} J_N} = 1.$$

The first column statement is theorem 6.4. □

7. **Reciprocal powers and parameter polynomials**

Since $\mathsf{D}_N(0) = 1$, its formal logarithm is defined at the origin. On a simply connected zero free region containing the origin, the same logarithm is holomorphic.

Definition 7.1. For $r \in \mathbb{C}$, define

$$\mathsf{S}_N^{(r)}(z) = \mathsf{D}_N(z)^{-r} = \exp\bigl(-r \log \mathsf{D}_N(z)\bigr).$$

Write

$$\mathsf{S}_N^{(r)}(z) = \sum_{n\geq 0} P_{N,n}(r) \frac{z^{Nn}}{(Nn)!}.$$

Define the logarithmic coefficients $\kappa_{N,m}$ by

$$-\log \Delta_N(w) = \sum_{m\geq 1} \kappa_{N,m} \frac{w^m}{m}.$$

The polynomial dependence on r can be read directly from the formal logarithm. We use the complete exponential Bell polynomials B_n defined by

$$\exp\left(\sum_{m\geq 1} x_m \frac{t^m}{m!}\right) = \sum_{n\geq 0} B_n(x_1, \ldots, x_n) \frac{t^n}{n!}.$$

THEOREM 7.2 (Bell polynomial formula). *For every $n \geq 0$,*

$$P_{N,n}(r) = \frac{(Nn)!}{n!} B_n\bigl(r\kappa_{N,1}, r1!\kappa_{N,2}, \ldots, r(n-1)!\kappa_{N,n}\bigr).$$

For $n \geq 1$, this is a polynomial in r of exact degree n. Its leading coefficient is

$$\frac{(Nn)!}{n!(N!)^n}.$$

Proof. The identity

$$\Delta_N(w)^{-r} = \exp\left(r \sum_{m\geq 1} \kappa_{N,m} \frac{w^m}{m}\right)$$

fits the defining generating function for B_n with

$$x_m = r(m-1)!\kappa_{N,m}.$$

The coefficient of w^n is therefore the displayed Bell polynomial divided by $n!$. Multiplication by $(Nn)!$ gives $P_{N,n}(r)$. Since

$$\Delta_N(w) = 1 - \frac{w}{N!} + O(w^2),$$

we have $\kappa_{N,1} = 1/N!$. The term x_1^n gives the stated leading coefficient, which is nonzero. □

The coefficients also satisfy a triangular recurrence that does not require the logarithmic coefficients to be known first.

THEOREM 7.3 (Parameter recurrence). *We have $P_{N,0}(r) = 1$. For $n \geq 1$,*

$$\begin{aligned} P_{N,n}(r) = & -\sum_{k=1}^{n-1}(-1)^k \binom{Nn-1}{Nk} P_{N,n-k}(r) \\ & - r\sum_{k=1}^{n}(-1)^k \binom{Nn-1}{Nk-1} P_{N,n-k}(r). \end{aligned}$$

Proof. Set $S = \mathsf{D}_N^{-r}$. Logarithmic differentiation gives

$$\mathsf{D}_N S' + r\mathsf{D}'_N S = 0.$$

Take the coefficient of $z^{Nn-1}/(Nn-1)!$. The term with $k = 0$ in the first product is $P_{N,n}(r)$. Every other term has a smaller block index. Rearrangement gives the formula. □

THEOREM 7.4 (Reciprocal semigroup). *For all $r, s \in \mathbb{C}$,*

$$\mathsf{S}_N^{(r)}(z)\mathsf{S}_N^{(s)}(z) = \mathsf{S}_N^{(r+s)}(z)$$

as a formal identity and on every common zero free region with a fixed logarithm. Consequently,

$$P_{N,n}(r+s) = \sum_{q=0}^{n} \binom{Nn}{Nq} P_{N,q}(r) P_{N,n-q}(s).$$

Proof. Both statements follow by adding the exponents of the same logarithm and then multiplying the block exponential generating series. □

COROLLARY 7.5 (A sequence of binomial type). *Define*

$$A_{N,n}(r) = \frac{n!}{(Nn)!} P_{N,n}(r).$$

Then

$$A_{N,n}(r+s) = \sum_{q=0}^{n} \binom{n}{q} A_{N,q}(r) A_{N,n-q}(s).$$

Thus $(A_{N,n})_{n\geq 0}$ is a polynomial sequence of binomial type.

Proof. Multiply the coefficient identity in theorem 7.4 by $n!/(Nn)!$ and simplify the factorials. □

The recurrence in theorem 7.3 uses the coefficients of the original denominator. A second recurrence uses only the logarithmic coefficients and is often shorter.

THEOREM 7.6 (Logarithmic Newton recurrence). *Set*

$$b_{N,n}(r) = \frac{P_{N,n}(r)}{(Nn)!}.$$

Then $b_{N,0}(r) = 1$ *and*

$$nb_{N,n}(r) = r\sum_{k=1}^{n} \kappa_{N,k} b_{N,n-k}(r).$$

Equivalently,

$$P_{N,n}(r) = \frac{r}{n}\sum_{k=1}^{n} \frac{(Nn)!}{(N(n-k))!}\kappa_{N,k} P_{N,n-k}(r).$$

If $p_k = r\kappa_{N,k}$, *then*

$$b_{N,n}(r) = \frac{1}{n!}\det\begin{pmatrix} p_1 & -1 & 0 & \cdots & 0 \\ p_2 & p_1 & -2 & \cdots & 0 \\ p_3 & p_2 & p_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -(n-1) \\ p_n & p_{n-1} & \cdots & p_2 & p_1 \end{pmatrix}.$$

Proof. Let

$$B_r(w) = \Delta_N(w)^{-r} = \sum_{n\geq 0} b_{N,n}(r) w^n.$$

Logarithmic differentiation gives

$$\frac{B_r'(w)}{B_r(w)} = r\sum_{k\geq 1} \kappa_{N,k} w^{k-1}.$$

Comparison of the coefficient of w^{n-1} proves the recurrence.

Let D_n denote the displayed determinant and put $D_0 = 1$. Expansion of this lower Hessenberg determinant gives

$$D_n = \sum_{k=1}^{n} \frac{(n-1)!}{(n-k)!} p_k D_{n-k}.$$

Thus $D_n/n!$ satisfies the same recurrence and initial value as $b_{N,n}(r)$. □

COROLLARY 7.7 (The first three reciprocal polynomials). *For every* $N \geq 1$,

$$P_{N,1}(r) = r,$$

$$P_{N,2}(r) = r\left[\frac{1}{2}\binom{2N}{N}(r+1) - 1\right],$$

and

$$P_{N,3}(r) = r\left[\frac{(3N)!}{6(N!)^3}(r+1)(r+2) - \binom{3N}{N}(r+1) + 1\right].$$

Proof. The first logarithmic coefficients are

$$\kappa_{N,1} = \frac{1}{N!},$$
$$\kappa_{N,2} = \frac{1}{(N!)^2} - \frac{2}{(2N)!},$$

and

$$\kappa_{N,3} = \frac{1}{(N!)^3} - \frac{3}{N!(2N)!} + \frac{3}{(3N)!}.$$

Insert these values into theorem 7.2 and simplify. □

COROLLARY 7.8. *For $n \geq 1$, the polynomial $P_{N,n}(r)$ is divisible by r.*

Proof. At $r = 0$, the reciprocal family is the constant function one. Every positive block coefficient therefore vanishes. □

Example 7.9. For $N = 1$, one has $\mathsf{D}_1(z)^{-r} = e^{rz}$ and $P_{1,n}(r) = r^n$. For $N = 2$,

$$P_{2,1}(r) = r,$$
$$P_{2,2}(r) = r(3r + 2),$$

and

$$P_{2,3}(r) = r(15r^2 + 30r + 16).$$

At $r = 1$, these are the positive secant coefficients.

8. **Logarithmic derivatives and Riccati equations**

Definition 8.1. Define the transfer tangent

$$\mathsf{T}_N(z) = -\frac{\mathsf{D}_N'(z)}{\mathsf{D}_N(z)}.$$

Its block expansion has the form

$$\mathsf{T}_N(z) = \sum_{n\geq 0} T_N(n) \frac{z^{Nn+N-1}}{(Nn+N-1)!}.$$

PROPOSITION 8.2 (Logarithmic coefficient formula). *The transfer tangent satisfies*

$$\mathsf{T}_N(z) = N \sum_{m\geq 1} \kappa_{N,m} z^{Nm-1}.$$

Equivalently,

$$T_N(n) = N(Nn + N - 1)!\kappa_{N,n+1}.$$

Proof. Since $\mathsf{D}_N(z) = \Delta_N(z^N)$,

$$-\frac{\mathsf{D}_N'(z)}{\mathsf{D}_N(z)} = -Nz^{N-1}\frac{\Delta_N'(z^N)}{\Delta_N(z^N)}.$$

Differentiate the defining series for $-\log \Delta_N$ and compare coefficients. □

THEOREM 8.3 (Tangent recurrence). *The coefficients satisfy* $T_N(0) = 1$ *and*

$$T_N(n) = (-1)^n - \sum_{k=1}^{n} (-1)^k \binom{Nn + N - 1}{Nk} T_N(n-k).$$

Proof. The defining equation is

$$\mathsf{D}_N \mathsf{T}_N = -\mathsf{D}'_N.$$

Extract the coefficient of $z^{Nn+N-1}/(Nn+N-1)!$. □

COROLLARY 8.4 (Derivative of the reciprocal family). *For every* $r \in \mathbb{C}$,

$$\frac{d}{dz} \mathsf{S}_N^{(r)}(z) = r \mathsf{T}_N(z) \mathsf{S}_N^{(r)}(z).$$

Proof. Differentiate D_N^{-r} on a zero free region. The formal identity follows from the same logarithmic calculation. □

For a function g, define the complete differential Bell polynomial Y_N by

$$\frac{d^N}{dz^N} e^{g(z)} = e^{g(z)} Y_N\big(g', g'', \dots, g^{(N)}\big).$$

THEOREM 8.5 (Higher Riccati equation). *On every zero free region of* D_N,

$$Y_N\big(-\mathsf{T}_N, -\mathsf{T}'_N, \dots, -\mathsf{T}_N^{(N-1)}\big) = -1.$$

Proof. Write $\mathsf{D}_N = e^g$ locally. Then $g' = -\mathsf{T}_N$. Divide the equation $\mathsf{D}_N^{(N)} = -\mathsf{D}_N$ by D_N and use the defining identity for Y_N. □

Example 8.6. For $N = 1$, $\mathsf{T}_1 = 1$. For $N = 2$, $\mathsf{D}_2 = \cos z$ and $\mathsf{T}_2 = \tan z$. The Riccati equation becomes

$$\mathsf{T}'_2 = 1 + \mathsf{T}_2^2.$$

9. **Congruential reciprocal functions and Appell systems**

The section $\mathsf{E}_{N,j}$ has a forced zero of order j at the origin. Removing it leaves a unit series in z^N.

Definition 9.1. For $0 \le j < N$, define

$$\Phi_{N,j}(w) = j! \sum_{n \ge 0} \frac{w^n}{(Nn + j)!}.$$

Then

$$\mathsf{E}_{N,j}(z) = \frac{z^j}{j!} \Phi_{N,j}(z^N).$$

Define the congruential reciprocal function

$$\mathsf{C}_{N,j}(z) = \Phi_{N,j}(z^N)^{-1} = \frac{z^j / j!}{\mathsf{E}_{N,j}(z)}.$$

Write

$$\mathsf{C}_{N,j}(z) = \sum_{n\geq 0} C_{N,j}(n)\frac{z^{Nn}}{(Nn)!}.$$

THEOREM 9.2 (Triangular inverse system). *The coefficients are determined by* $C_{N,j}(0) = 1$ *and*

$$\sum_{q=0}^{n} \binom{Nn+j}{Nq} C_{N,j}(q) = \delta_{n0}.$$

For $n \geq 1$,

$$C_{N,j}(n) = -\frac{\displaystyle\sum_{q=0}^{n-1} \binom{Nn+j}{Nq} C_{N,j}(q)}{\displaystyle\binom{Nn+j}{Nn}}.$$

Proof. Multiply the defining series for $\mathsf{C}_{N,j}$ and $\mathsf{E}_{N,j}$. Their product is $z^j/j!$. At degree $Nn+j$, multiplication by $(Nn+j)!$ gives the stated binomial coefficient. The diagonal coefficient is nonzero, so the system is triangular. □

Definition 9.3. For $r \in \mathbb{C}$, define formal coefficients $C_{N,j,n}(r)$ by

$$\mathsf{C}_{N,j}(z)^r = \sum_{n\geq 0} C_{N,j,n}(r)\frac{z^{Nn}}{(Nn)!}.$$

On a simply connected region that avoids the zeros of $\Phi_{N,j}(z^N)$, the same definition uses the holomorphic and nonvanishing function $\mathsf{C}_{N,j}$ together with one fixed logarithm.

THEOREM 9.4 (Congruential reciprocal semigroup). *For each* $n \geq 1$, *the coefficient* $C_{N,j,n}(r)$ *is a polynomial in* r *of exact degree* n. *Its leading coefficient is*

$$\frac{(Nn)!}{n!}\left(-\frac{j!}{(N+j)!}\right)^n.$$

It satisfies

$$C_{N,j,n}(r+s) = \sum_{q=0}^{n} \binom{Nn}{Nq} C_{N,j,q}(r) C_{N,j,n-q}(s).$$

In particular,

$$C_{N,j,n}(1) = C_{N,j}(n).$$

Proof. The unit series $\Phi_{N,j}(w)^{-1}$ has a formal logarithm with zero constant term. Its linear term is

$$-\frac{j!}{(N+j)!}w.$$

Exponentiation after multiplication by r shows that the coefficient of w^n is a polynomial of degree at most n. The nth power of the linear logarithmic term gives the displayed nonzero leading coefficient, so the degree is exactly n. Multiplication of the powers with parameters r and s gives the coefficient identity. □

Definition 9.5. Define polynomials $E_m^{(N,j,r)}(x)$ by

$$\mathsf{C}_{N,j}(z)^r e^{xz} = \sum_{m\geq 0} E_m^{(N,j,r)}(x)\frac{z^m}{m!}.$$

Theorem 9.6 (Appell laws). *The polynomials satisfy*

$$\frac{d}{dx}E_m^{(N,j,r)}(x) = mE_{m-1}^{(N,j,r)}(x)$$

and

$$E_m^{(N,j,r)}(x+y) = \sum_{q=0}^{m}\binom{m}{q}E_q^{(N,j,r)}(x)y^{m-q}.$$

Proof. Differentiate the generating function with respect to x for the first identity. Multiply it by e^{yz} and compare coefficients for the second. □

Proposition 9.7 (Explicit Appell polynomial). *For every $m \geq 0$,*

$$E_m^{(N,j,r)}(x) = \sum_{0\leq n\leq m/N}\binom{m}{Nn}C_{N,j,n}(r)x^{m-Nn}.$$

Proof. Multiply the block series for $\mathsf{C}_{N,j}^r$ by the exponential series and collect the coefficient of total degree m. □

Theorem 9.8 (Two parameter Appell convolution). *For all $r, s, x, y \in \mathbb{C}$ and $m \geq 0$,*

$$E_m^{(N,j,r+s)}(x+y) = \sum_{q=0}^{m}\binom{m}{q}E_q^{(N,j,r)}(x)E_{m-q}^{(N,j,s)}(y).$$

Proof. Multiply the generating functions

$$\mathsf{C}_{N,j}(z)^r e^{xz} \quad \text{and} \quad \mathsf{C}_{N,j}(z)^s e^{yz}.$$

Their product is the generating function with parameters $r+s$ and $x+y$. Coefficient comparison gives the formula. □

Proposition 9.9 (Differential operator form). *Let a formal series $A(z) = \sum_{q\geq 0} a_q z^q/q!$ act on polynomials by*

$$A(\partial_x)f(x) = \sum_{q\geq 0}\frac{a_q}{q!}\partial_x^q f(x).$$

The power r is applied to the formal symbol before the substitution $z = \partial_x$. Then

$$E_m^{(N,j,r)}(x) = \mathsf{C}_{N,j}(\partial_x)^r x^m.$$

The inverse relation is

$$x^m = \Phi_{N,j}(\partial_x)^r E_m^{(N,j,r)}(x).$$

Both sums are finite on every polynomial.

Proof. Applying $A(\partial_x)$ to e^{xz} multiplies it by $A(z)$. Taking $A = \mathsf{C}_{N,j}^r$ gives the first identity by the defining generating function. The operators defined by $\mathsf{C}_{N,j}^r$ and $\Phi_{N,j}^r$ are inverse because their symbols multiply to one. □

Corollary 9.10 (Appell symmetry). *Each polynomial $E_m^{(N,j,r)}$ is monic of degree m, and*

$$E_m^{(N,j,r)}(\omega x) = \omega^m E_m^{(N,j,r)}(x).$$

Proof. The term $n = 0$ in proposition 9.7 is x^m. Every other exponent is $m - Nn$, so replacement of x by ωx multiplies every term by ω^m. □

Example 9.11. For $(N, j) = (2, 0)$,

$$\mathsf{C}_{2,0}(z) = \operatorname{sech} z.$$

For $(N, j) = (2, 1)$,

$$\mathsf{C}_{2,1}(z) = \frac{z}{\sinh z}.$$

The first case is a scaled and translated Euler family. In fact, if $E_m(x)$ denotes the classical Euler polynomial, then

$$E_m^{(2,0,1)}(x) = 2^m E_m\left(\frac{x+1}{2}\right).$$

The second case belongs to the familiar Bernoulli type family.

10. **Scaled reciprocal coupling and theta polynomials**

Definition 10.1. Set

$$H_N(z) = \mathsf{E}_{N,0}(z) = \sum_{n\geq 0} \frac{z^{Nn}}{(Nn)!}.$$

For $u \in \mathbb{C}$, define $\Theta_{N,n}(u)$ by

$$H_N(uz)^{-1} H_N(z)^{-u} = \sum_{n\geq 0} \Theta_{N,n}(u) \frac{z^{Nn}}{(Nn)!}.$$

THEOREM 10.2 (Theta reconstruction). *For every $n \geq 0$,*

$$\Theta_{N,n}(u) = (-1)^n \sum_{q=0}^{n} \binom{Nn}{Nq} P_{N,q}(1) u^{Nq} P_{N,n-q}(u).$$

Thus $\Theta_{N,n}$ is a polynomial of degree at most Nn.

Proof. Choose ρ with $\rho^N = -1$. Then

$$H_N(z) = \mathsf{D}_N(\rho z).$$

The coefficient of z^{Nn} in $H_N(z)^{-r}$ is therefore $(-1)^n P_{N,n}(r)/(Nn)!$. Apply this to $H_N(uz)^{-1}$ and $H_N(z)^{-u}$, then multiply the two block series. □

PROPOSITION 10.3 (Trivial parameter factors). *For $n \geq 1$, the polynomial $\Theta_{N,n}(u)$ is divisible by u. If N is even, it is also divisible by $u+1$.*

Proof. At $u = 0$, the defining product is one. Hence every positive coefficient vanishes. If N is even, then $H_N(-z) = H_N(z)$. At $u = -1$, the product is again one, so every positive coefficient vanishes there as well. □

PROPOSITION 10.4 (Theta Newton recurrence). *Put*

$$\theta_{N,n}(u) = \frac{\Theta_{N,n}(u)}{(Nn)!}.$$

Then $\theta_{N,0}(u) = 1$ and

$$n\theta_{N,n}(u) = \sum_{m=1}^{n} (-1)^m \kappa_{N,m} \big(u^{Nm} + u\big) \theta_{N,n-m}(u).$$

In particular,

$$\Theta_{N,1}(u) = -(u^N + u).$$

Proof. Write $w = z^N$. Since $H_N(z) = \Delta_N(-w)$,

$$-\log H_N(z) = \sum_{m \geq 1} (-1)^m \kappa_{N,m} \frac{w^m}{m}.$$

The logarithm of the theta generating function is therefore

$$\sum_{m \geq 1} (-1)^m \kappa_{N,m} \big(u^{Nm} + u\big) \frac{w^m}{m}.$$

Logarithmic differentiation and coefficient comparison give the recurrence. The case $n = 1$ uses $\kappa_{N,1} = 1/N!$. □

Example 10.5. For $N = 2$,

$$\Theta_{2,1}(u) = -u(u+1)$$

and

$$\Theta_{2,2}(u) = u(u+1)(5u^2 + u + 2).$$

The two factors predicted by proposition 10.3 are visible.

11. **Growth and the geometry of the large zeros**

We now assume $N \geq 2$. Write

$$\eta_k = e^{(2k+1)\pi i/N}, \qquad 0 \leq k < N.$$

These are the vertices of a regular polygon and

$$\mathsf{D}_N(z) = \frac{1}{N} \sum_{k=0}^{N-1} e^{\eta_k z}.$$

Standard Fact 11.1 (Order from Taylor coefficients). If $f(w) = \sum_{n\geq 0} a_n w^n$ is an entire function with infinitely many nonzero coefficients, then

$$\operatorname{ord} f = \limsup_{n\to\infty} \frac{n \log n}{\log(1/|a_n|)}.$$

Canonical products for entire functions of order below one have genus zero. We use these standard facts in the form given in [Boa54] and [Tit39].

PROPOSITION 11.2 (Orders and exponential type). *For $N \geq 2$,*

$$\operatorname{ord} \Delta_N = \frac{1}{N}.$$

The function D_N has order one and exponential type one.

Proof. The coefficient of w^n in Δ_N has modulus $1/(Nn)!$. Stirling's formula and standard fact 11.1 give order $1/N$. Composition with z^N gives order one for D_N.

The finite exponential representation gives

$$|\mathsf{D}_N(z)| \leq e^{|z|},$$

so the type is at most one. Choose a root η_0 and put $z = t\overline{\eta_0}$ with $t > 0$. The term $e^{\eta_0 z} = e^t$ has strictly larger real exponent than every other term. Hence

$$\mathsf{D}_N(t\overline{\eta_0}) = \frac{1}{N} e^t (1 + o(1)).$$

The type is therefore one. □

Definition 11.3. For $\theta \in \mathbb{R}$, define

$$h_N(\theta) = \max_{0\leq k<N} \cos\left(\theta + \frac{(2k+1)\pi}{N}\right).$$

The tie directions are

$$\mathfrak{T}_N = \left\{ \frac{2\pi k}{N} : 0 \leq k < N \right\} \mod 2\pi.$$

THEOREM 11.4 (Indicator polygon and dominant sectors). *The Phragmén Lindelöf indicator of* D_N *is* h_N. *More precisely,*

$$\limsup_{r\to\infty} \frac{1}{r} \log |\mathsf{D}_N(re^{i\theta})| = h_N(\theta).$$

If K *is a compact set of directions disjoint from* $\mathfrak{T}_N$, *then there are* $\delta > 0$ *and* $R > 0$ *such that, for each* $\theta \in K$, *one root* $\eta(\theta)$ *of* -1 *satisfies*

$$\mathsf{D}_N(re^{i\theta}) = \frac{1}{N} \exp\big(\eta(\theta) re^{i\theta}\big) \left(1 + O(e^{-\delta r})\right)$$

uniformly for $r \geq R$. *In particular, no sufficiently large zero can lie in those directions.*

Proof. For $z = re^{i\theta}$, the real part of the exponent $\eta_k z$ is

$$r \cos\left(\theta + \frac{(2k+1)\pi}{N}\right).$$

The triangle inequality gives the upper indicator bound $h_N(\theta)$.

Away from a tie direction, the maximizing vertex is unique. On a compact set avoiding all ties, the difference between its projection and every other projection has a uniform positive lower bound. Division by the maximizing exponential gives the stated uniform asymptotic and the zero free conclusion.

At a tie direction exactly two adjacent vertices attain the maximum. After rotation it is enough to take $\theta = 0$. The two dominant terms give

$$\frac{2}{N} e^{c_N r} \cos(s_N r)$$

and every other term is exponentially smaller. Along a sequence on which $|\cos(s_N r)| \geq 1/2$, the lower indicator bound is $c_N = h_N(0)$. Rotational symmetry treats every tie direction. □

The zero strings are governed by two adjacent exponents. Put

$$a_N = \frac{\pi}{N}, \qquad s_N = \sin a_N, \qquad c_N = \cos a_N,$$

and

$$A_N = \frac{\pi}{s_N}, \qquad t_m = A_N \left(m + \frac{1}{2}\right).$$

For $N \geq 3$, also put

$$\gamma_N = c_N - \cos(3a_N) > 0.$$

LEMMA 11.5 (Two term dominance). *Assume* $N \geq 3$. *There are positive constants* Y, R, *and* C *with the following properties.*

If $z = x + iy$ *lies in the sector* $|\arg z| \leq a_N$, *has* $x \geq R$, *and is a zero of* D_N, *then* $|y| \leq Y$.

On the strip $x \geq R$ and $|y| \leq Y$, define

$$R_N(z) = \sum_{\substack{\eta^N=-1 \\ \eta \neq e^{ia_N}, e^{-ia_N}}} \exp\big((\eta - e^{-ia_N})z\big).$$

Then

$$Ne^{-e^{-ia_N}z}\mathsf{D}_N(z) = 1 + e^{2is_N z} + R_N(z),$$

and, for $q = 0, 1, 2$,

$$|R_N^{(q)}(z)| \leq Ce^{-\gamma_N x}.$$

Proof. The roots e^{ia_N} and e^{-ia_N} have the largest real parts on the positive ray. Every other root has real part at most $\cos(3a_N)$. On a fixed horizontal strip, division by $e^{e^{-ia_N}z}$ gives the displayed identity. Differentiating a remainder term only multiplies it by a bounded constant. The bounded imaginary part contributes a fixed factor, so the estimates hold for $q = 0, 1, 2$.

We now exclude zeros outside a fixed strip. By theorem 11.4, directions separated from the positive ray are eventually zero free. It remains to consider a narrow cone $|y| \leq \varepsilon x$. Choose $\varepsilon > 0$ so small that every exponent other than e^{ia_N} and e^{-ia_N} remains smaller than both dominant candidates by a factor $e^{-\delta x}$ for some $\delta > 0$.

If $y \geq Y$, the modulus of the term with exponent e^{-ia_N} exceeds that of the term with exponent e^{ia_N} by $e^{2s_N y}$. Choose Y so that this ratio is larger than four. For sufficiently large x, the remaining terms have total modulus less than one fourth of the largest term. Cancellation is then impossible. Conjugation gives the same conclusion for $y \leq -Y$. □

Lemma 11.6 (Near cancellation of two exponentials). *For every $M > 0$, there are constants $\varepsilon_M > 0$ and $C_M > 0$ such that the following holds. If*

$$|\Re w| \leq M$$

and

$$|1 + e^w| \leq \varepsilon_M,$$

then there is a unique integer m with

$$|w - (2m + 1)\pi i| \leq C_M|1 + e^w|.$$

Proof. The function $1 + e^w$ is $2\pi i$ periodic and has simple zeros at the odd multiples of πi. Reduce the imaginary part to a compact interval of length 2π. Away from small disks about the zeros, the modulus has a positive minimum. Inside each disk, the inverse function theorem gives the stated linear estimate. Shrinking ε_M makes the relevant zero unique. □

THEOREM 11.7 (Large zero strings). *Assume $N \geq 3$. There are constants m_0, C, and $\gamma > 0$ and a sequence of positive real simple zeros $(\tau_m)_{m \geq m_0}$ of D_N such that*

$$\tau_m = t_m + O(e^{-\gamma t_m}).$$

Every zero of sufficiently large modulus is of the form

$$\omega^k \tau_m$$

for a unique $0 \leq k < N$ and a unique $m \geq m_0$.

The derivatives satisfy

$$\mathsf{D}_N'(\omega^k \tau_m) = \omega^{-k} \mathsf{D}_N'(\tau_m)$$

and

$$|\mathsf{D}_N'(\tau_m)| = \frac{2s_N}{N} e^{c_N \tau_m} \left(1 + O(e^{-\gamma \tau_m})\right).$$

Proof. Work first in the sector $|\arg z| \leq a_N$. By lemma 11.5, every sufficiently large zero lies in a fixed strip about the positive real axis. Put

$$G(z) = N e^{-e^{-i a_N} z} \mathsf{D}_N(z).$$

Then

$$G(z) = 1 + e^{2i s_N z} + R_N(z)$$

with an exponentially small remainder and derivative.

If $G(z) = 0$, then

$$|1 + e^{2i s_N z}| \leq C e^{-\gamma_N \Re z}.$$

The real part of $2i s_N z$ is bounded in the strip. By lemma 11.6, the point z lies at distance $O(e^{-\gamma_N \Re z})$ from one of the real points

$$t_m = \frac{(2m+1)\pi}{2s_N}.$$

Fix a large m. On a circle

$$|z - t_m| = K e^{-\gamma_N t_m}$$

with K fixed and large, the function $1 + e^{2i s_N z}$ has modulus at least a constant times $K e^{-\gamma_N t_m}$. The remainder is at most a smaller constant times $e^{-\gamma_N t_m}$. Rouché's theorem gives exactly one zero of G in the disk. The count is with multiplicity, so this zero is simple. The disk is invariant under complex conjugation and D_N has real coefficients. Uniqueness forces the zero to be real. Denote it by τ_m.

The localization argument shows that these disks contain every large zero in the fundamental sector. Rotational invariance gives the zeros $\omega^k \tau_m$ and covers the plane. The disks are disjoint for large m, which gives uniqueness of k and m.

Differentiate the normalized expression. At $z = \tau_m$,

$$G'(\tau_m) = 2is_N e^{2is_N\tau_m} + R_N'(\tau_m) = -2is_N + O(e^{-\gamma\tau_m}).$$

Since $G(\tau_m) = 0$,

$$\mathsf{D}_N'(\tau_m) = \frac{1}{N} e^{e^{-ia_N\tau_m}\tau_m} G'(\tau_m).$$

Taking absolute values gives the derivative asymptotic. Differentiating $\mathsf{D}_N(\omega z) = \mathsf{D}_N(z)$ gives the rotation formula. □

THEOREM 11.8 (First displacement and derivative sign). *With the notation of lemma* 11.5 *and theorem* 11.7,

$$\tau_m = t_m + \frac{R_N(t_m)}{2is_N} + O(e^{-2\gamma_N t_m}).$$

The quotient $R_N(t_m)/(2is_N)$ *is real. One also has*

$$\mathsf{D}_N'(\tau_m) = (-1)^{m+1}\frac{2s_N}{N} e^{c_N\tau_m}\left(1 + O(e^{-\gamma_N\tau_m})\right).$$

Proof. Put $\delta_m = \tau_m - t_m$. The large zero theorem gives $\delta_m = O(e^{-\gamma_N t_m})$. Since $e^{2is_N t_m} = -1$,

$$1 + e^{2is_N(t_m+\delta_m)} = -2is_N\delta_m + O(\delta_m^2).$$

The remainder estimate through the first derivative gives

$$R_N(t_m + \delta_m) = R_N(t_m) + O(e^{-\gamma_N t_m}|\delta_m|).$$

Substitution in the zero equation gives the displacement formula.

At the real point t_m, the normalized value

$$Ne^{-e^{-ia_N t_m}}\mathsf{D}_N(t_m)$$

is purely imaginary. The two dominant terms cancel there, so $R_N(t_m)$ is purely imaginary and the displayed correction is real.

For the derivative, use the formula in the proof of theorem 11.7 and

$$e^{-is_N\tau_m} = (-1)^m(-i)\left(1 + O(e^{-\gamma_N\tau_m})\right).$$

The stated real sign follows. □

COROLLARY 11.9 (The first correction for $N = 3$). *For* $N = 3$,

$$t_m = \frac{2\pi}{\sqrt{3}}\left(m + \frac{1}{2}\right)$$

and

$$\tau_m = t_m + \frac{(-1)^m}{\sqrt{3}} e^{-3t_m/2} + O(e^{-3t_m}).$$

Proof. The only nondominant exponent is -1. Hence

$$R_3(t) = \exp\left(\left(-\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)t\right).$$

At $t = t_m$, its phase is $i(-1)^m$. Insert this value into theorem 11.8. □

COROLLARY 11.10 (The case $N = 2$). *For $N = 2$,*

$$\mathsf{D}_2(z) = \cos z.$$

Its zeros are

$$\pm\left(m + \frac{1}{2}\right)\pi, \qquad m \geq 0,$$

and all are simple. They agree with the formula $t_m = A_2(m + 1/2)$.

Proof. This is the standard zero set of the cosine function. □

COROLLARY 11.11 (Eventual simplicity and exact rays). *For every $N \geq 2$, all but finitely many zeros of D_N are simple. Every sufficiently large zero lies on one of the rays*

$$\omega^k(0, \infty), \qquad 0 \leq k < N.$$

Proof. Use theorem 11.7 for $N \geq 3$ and corollary 11.10 for $N = 2$. □

Example 11.12 (Exact fourth order strings). For $N = 4$,

$$\mathsf{D}_4(z) = \cos\left(\frac{z}{\sqrt{2}}\right)\cosh\left(\frac{z}{\sqrt{2}}\right).$$

The positive zeros are exactly

$$\tau_m = \sqrt{2}\pi\left(m + \frac{1}{2}\right) = A_4\left(m + \frac{1}{2}\right).$$

Rotation by the fourth roots of unity gives the real and imaginary zero strings. In this case the error term in theorem 11.7 is identically zero.

COROLLARY 11.13 (Reduced zeros). *Let Λ_N be the nonzero zero multiset of Δ_N. All but finitely many members of Λ_N are positive and simple. They can be indexed as*

$$\lambda_m = \tau_m^N$$

for $m \geq m_0$, and

$$\lambda_m = t_m^N + O\left(t_m^{N-1} e^{-\gamma t_m}\right).$$

Proof. The identity $\mathsf{D}_N(z) = \Delta_N(z^N)$ identifies each orbit of N nonzero zeros of D_N with one zero of Δ_N. The large orbit representatives are the positive zeros τ_m. Simplicity is preserved because

$$\mathsf{D}_N'(z) = N z^{N-1} \Delta_N'(z^N)$$

and $z \neq 0$. The asymptotic follows by expanding τ_m^N about t_m. □

THEOREM 11.14 (Zero counting asymptotics). *Let $n_{\mathsf{D}_N}(R)$ count the zeros of D_N in $|z| \le R$, with multiplicity. Let $n_{\Delta_N}(R)$ count the zeros of Δ_N in $|w| \le R$. Then*

$$n_{\mathsf{D}_N}(R) = \frac{N \sin(\pi/N)}{\pi} R + O(1)$$

and

$$n_{\Delta_N}(R) = \frac{\sin(\pi/N)}{\pi} R^{1/N} + O(1).$$

Proof. Each of the N rays contains one large zero near

$$t_m = \frac{\pi}{\sin(\pi/N)} \left(m + \frac{1}{2} \right).$$

The number of such indices with $t_m \le R$ is

$$\frac{\sin(\pi/N)}{\pi} R + O(1).$$

Multiplication by N and addition of the finite exceptional set give the first formula. The second follows from

$$n_{\mathsf{D}_N}(R^{1/N}) = N n_{\Delta_N}(R).$$

□

COROLLARY 11.15 (Zero exponents). *The exponent of convergence of the nonzero zeros of D_N is one. The exponent of convergence of the zeros of Δ_N is $1/N$.*

Proof. The exact counting laws in theorem 11.14 give the two exponents by partial summation. □

12. **Hadamard products and zero power sums**

The reduction from z to $w = z^N$ removes the rotational repetition of the zeros.

THEOREM 12.1 (Genus zero product). *For $N \ge 2$, let Λ_N be the nonzero zero multiset of Δ_N. Then*

$$\Delta_N(w) = \prod_{\lambda \in \Lambda_N} \left(1 - \frac{w}{\lambda} \right),$$

where the product converges locally uniformly. Consequently,

$$\mathsf{D}_N(z) = \prod_{\lambda \in \Lambda_N} \left(1 - \frac{z^N}{\lambda} \right).$$

Proof. By proposition 11.2, the order of Δ_N is $1/N < 1$. By corollary 11.13, it has infinitely many zeros. Hadamard factorization therefore gives a canonical product of genus zero. The only possible exponential factor is a nonzero constant because the order is below one. Since $\Delta_N(0) = 1$, that constant is one. Substitution of $w = z^N$ gives the second formula. □

THEOREM 12.2 (Trace class determinant model). *Let $\mathcal{H}_N$ have an orthonormal basis indexed by the members of Λ_N, with multiplicity. Define the diagonal operator*

$$K_N e_\lambda = \lambda^{-1} e_\lambda.$$

Then K_N is trace class and

$$\Delta_N(w) = \det_F(I - wK_N).$$

For every $m \geq 1$,

$$\kappa_{N,m} = \operatorname{tr}(K_N^m).$$

Proof. By corollary 11.15,

$$\sum_{\lambda \in \Lambda_N} |\lambda|^{-1} < \infty.$$

Thus the diagonal operator K_N is trace class. Its Fredholm determinant is the locally uniform product

$$\det_F(I - wK_N) = \prod_{\lambda \in \Lambda_N} \left(1 - \frac{w}{\lambda}\right),$$

which equals $\Delta_N(w)$ by theorem 12.1.

Taking the logarithm gives

$$-\log \Delta_N(w) = \sum_{m \geq 1} \frac{w^m}{m} \operatorname{tr}(K_N^m)$$

near the origin. Comparison with the definition of $\kappa_{N,m}$ proves the trace identity. □

COROLLARY 12.3 (Complete symmetric function form). *For a positive integer r, let each eigenvalue λ^{-1} of K_N be repeated r times. If h_n denotes the complete homogeneous symmetric function of degree n in this absolutely summable multiset, then*

$$\frac{P_{N,n}(r)}{(Nn)!} = h_n.$$

For arbitrary complex r, the Bell polynomial formula is the polynomial continuation obtained by multiplying every power sum by r.

Proof. The generating function for the complete homogeneous symmetric functions is

$$\prod_{\lambda\in\Lambda_N}\Big(1-\frac{w}{\lambda}\Big)^{-r}=\Delta_N(w)^{-r}.$$

Coefficient comparison gives the first identity. Power sums determine the complete homogeneous functions through the Newton recurrence. Replacing each power sum by r times itself gives theorems 7.2 and 7.6. □

PROPOSITION 12.4 (Products for the normalized sections). *For every* $0\le j<N$, *the entire function*

$$\Phi_{N,j}(w)=j!\sum_{n\ge0}\frac{w^n}{(Nn+j)!}$$

has order $1/N$. *If* $N\ge2$ *and* $\Lambda_{N,j}$ *is its nonzero zero multiset, then*

$$\Phi_{N,j}(w)=\prod_{\mu\in\Lambda_{N,j}}\Big(1-\frac{w}{\mu}\Big).$$

For $m\ge1$, *put*

$$\sigma_{N,j,m}=\sum_{\mu\in\Lambda_{N,j}}\mu^{-m}.$$

Then

$$C_{N,j,n}(r)=\frac{(Nn)!}{n!}B_n\big(r\sigma_{N,j,1},r1!\sigma_{N,j,2},\dots,r(n-1)!\sigma_{N,j,n}\big).$$

Proof. The coefficient formula for the order gives $1/N$. The function is not a polynomial. An entire function of order below one with only finitely many zeros would be a polynomial times a constant by Hadamard factorization, which is impossible here. The genus zero product follows as in theorem 12.1.

Taking the logarithm gives

$$-\log\Phi_{N,j}(w)=\sum_{m\ge1}\sigma_{N,j,m}\frac{w^m}{m}.$$

Since $\mathsf{C}_{N,j}(z)^r=\Phi_{N,j}(z^N)^{-r}$, the Bell polynomial calculation from theorem 7.2 gives the coefficient formula. □

Definition 12.5. For $m\ge1$, define

$$\sigma_{N,m}=\sum_{\lambda\in\Lambda_N}\lambda^{-m},$$

with multiplicities.

The series converges absolutely because the exponent of convergence of Λ_N is $1/N<1$.

THEOREM 12.6 (Newton sums for the reduced zeros). *For $|w|$ below the least zero modulus*,

$$-\log \Delta_N(w) = \sum_{m\geq 1} \sigma_{N,m} \frac{w^m}{m}.$$

Thus

$$\kappa_{N,m} = \sigma_{N,m}.$$

If

$$a_m = \frac{(-1)^m}{(Nm)!}, \qquad a_0 = 1,$$

then

$$\sigma_{N,m} = -m a_m - \sum_{k=1}^{m-1} a_k \sigma_{N,m-k}.$$

Proof. Take the logarithm of the product in theorem 12.1. Absolute convergence in a disk below the first zero permits termwise expansion

$$-\log\left(1 - \frac{w}{\lambda}\right) = \sum_{m\geq 1} \frac{w^m}{m\lambda^m}.$$

This gives the logarithmic formula and identifies $\kappa_{N,m}$ with $\sigma_{N,m}$.

Differentiate the product identity in logarithmic form to obtain

$$\Delta_N'(w) = -\Delta_N(w) \sum_{m\geq 1} \sigma_{N,m} w^{m-1}.$$

Comparison of the coefficient of w^{m-1} gives

$$m a_m = -\sum_{k=0}^{m-1} a_k \sigma_{N,m-k}.$$

Solving for $\sigma_{N,m}$ gives the recurrence. □

COROLLARY 12.7 (Mittag Leffler form of the transfer tangent). *Away from the zeros of* D_N,

$$\mathsf{T}_N(z) = N z^{N-1} \sum_{\lambda\in\Lambda_N} \frac{1}{\lambda - z^N}.$$

The series converges locally uniformly away from the poles.

Proof. Logarithmic differentiation of the genus zero product gives

$$-\frac{\Delta_N'(w)}{\Delta_N(w)} = \sum_{\lambda\in\Lambda_N} \frac{1}{\lambda - w}.$$

The series is locally uniform because $\sum_\lambda |\lambda|^{-1} < \infty$. Substitute $w = z^N$ and use the chain rule. □

PROPOSITION 12.8 (Theta Bell polynomial formula). *Put*

$$\tau_{N,m} = (-1)^m \sigma_{N,m}.$$

Then

$$\Theta_{N,n}(u) = \frac{(Nn)!}{n!} B_n\big(\tau_{N,1}(u^N+u), 1!\tau_{N,2}(u^{2N}+u), \dots, (n-1)!\tau_{N,n}(u^{nN}+u)\big).$$

Proof. The reduced unsigned denominator is

$$\Psi_N(w) = \sum_{n\geq 0} \frac{w^n}{(Nn)!} = \Delta_N(-w).$$

By theorem 12.6,

$$-\log \Psi_N(w) = \sum_{m\geq 1} \tau_{N,m} \frac{w^m}{m}.$$

With $w = z^N$, the logarithm of the theta generating function is

$$\sum_{m\geq 1} \tau_{N,m}(u^{Nm}+u)\frac{w^m}{m}.$$

The complete Bell polynomial formula follows. □

COROLLARY 12.9 (Tangent coefficients as zero sums). *For every* $n \geq 0$,

$$T_N(n) = N(Nn+N-1)!\sigma_{N,n+1}.$$

Proof. Combine proposition 8.2 and theorem 12.6. □

COROLLARY 12.10 (Reciprocal coefficients from zero data). *For every* $n \geq 0$,

$$P_{N,n}(r) = \frac{(Nn)!}{n!} B_n\big(r\sigma_{N,1}, r1!\sigma_{N,2}, \dots, r(n-1)!\sigma_{N,n}\big).$$

Proof. Use theorems 7.2 and 12.6. □

Example 12.11 (The cosine orbit sums). For $N = 2$,

$$\Delta_2(w) = \cos\sqrt{w}.$$

Its zeros are

$$\lambda_m = \pi^2\left(m+\frac{1}{2}\right)^2, \qquad m \geq 0.$$

Hence

$$\sigma_{2,q} = \frac{2^{2q}-1}{\pi^{2q}}\zeta(2q).$$

The identity in corollary 12.9 recovers the classical tangent coefficients from these zero power sums.

13. **Coefficient asymptotics from the nearest zeros**

For an entire function f, let $\mathcal{Z}(f)$ denote its multiset of nonzero zeros. Let r be a positive integer. Then D_N^{-r} is meromorphic and a simple zero of D_N becomes a pole of order r.

THEOREM 13.1 (Full nearest circle expansion). *Suppose the zeros of D_N of least modulus R are simple and form the finite set $\mathcal{Z}_R$. Suppose the next zero modulus is $R_2 > R$. At each $\rho \in \mathcal{Z}_R$, write the principal part*

$$\mathsf{D}_N(z)^{-r} = \sum_{\ell=1}^{r} a_{\rho,\ell} \left(1 - \frac{z}{\rho}\right)^{-\ell} + O(1).$$

Then, for every R_* with $R < R_* < R_2$ and $m = Nn$,

$$P_{N,n}(r) = m! \sum_{\rho \in \mathcal{Z}_R} \sum_{\ell=1}^{r} a_{\rho,\ell} \binom{m+\ell-1}{\ell-1} \rho^{-m} + O_{R_*}(m! R_*^{-m}).$$

The leading Laurent coefficient is

$$a_{\rho,r} = (-\rho \mathsf{D}_N'(\rho))^{-r}.$$

Proof. Subtract the complete principal parts at the zeros on $|z| = R$. The remainder is holomorphic in every closed disk of radius $R_* < R_2$. Cauchy's estimate gives an ordinary coefficient bound $O_{R_*}(R_*^{-m})$. The coefficient of z^m in

$$\left(1 - \frac{z}{\rho}\right)^{-\ell}$$

is

$$\binom{m+\ell-1}{\ell-1} \rho^{-m}.$$

Multiplication by $m!$ converts the ordinary coefficient to $P_{N,n}(r)$.

Near a simple zero,

$$\mathsf{D}_N(z) = -\rho \mathsf{D}_N'(\rho) \left(1 - \frac{z}{\rho}\right) + O\left(\left(1 - \frac{z}{\rho}\right)^2\right).$$

Raising to the power $-r$ gives the formula for $a_{\rho,r}$. □

COROLLARY 13.2 (Leading dominant zero term). *Under the hypotheses of theorem* 13.1,

$$P_{N,n}(r) = m! \binom{m+r-1}{r-1} \sum_{\rho \in \mathcal{Z}_R} \frac{\rho^{-m}}{(-\rho \mathsf{D}_N'(\rho))^r} + O\left(m! m^{r-2} R^{-m}\right) + O_{R_*}\left(m! R_*^{-m}\right).$$

For $r = 1$, the first error term is absent.

Proof. The terms with $\ell < r$ in theorem 13.1 are bounded by a constant times $m^{r-2}R^{-m}$. The term with $\ell = r$ gives the displayed main term. □

Corollary 13.3 (Orbit reduced nearest circle term). *Assume the hypotheses of theorem* 13.1. *Choose one representative from each orbit of the nearest zeros under multiplication by* ω, *and call the resulting set* $\mathcal{Z}_R^{\mathrm{orb}}$. *With* $m = Nn$,

$$P_{N,n}(r) = Nm!\binom{m+r-1}{r-1}\sum_{\tau\in\mathcal{Z}_R^{\mathrm{orb}}}\frac{\tau^{-m}}{(-\tau\mathsf{D}'_N(\tau))^r} + O\left(m!m^{r-2}R^{-m}\right) + O_{R_*}\left(m!R_*^{-m}\right).$$

For $r = 1$, *the first error term is absent.*

Proof. If $\rho = \omega^k\tau$, then

$$\mathsf{D}'_N(\rho) = \omega^{-k}\mathsf{D}'_N(\tau).$$

Therefore

$$-\rho\mathsf{D}'_N(\rho) = -\tau\mathsf{D}'_N(\tau).$$

Since $m = Nn$, one also has $\rho^{-m} = \tau^{-m}$. Every orbit therefore contributes N equal terms in corollary 13.2. □

Corollary 13.4 (Secant asymptotic). *For* $N = 2$ *and* $r = 1$,

$$P_{2,n}(1) \sim 2(2n)!\left(\frac{2}{\pi}\right)^{2n+1}.$$

Proof. The nearest zeros of $\cos z$ are $\rho = \pm\pi/2$, both simple. Their contributions in corollary 13.2 are equal. □

14. **Zero zeta functions**

The zero counting theorem gives the exact abscissae of absolute convergence. The exponentially accurate zero locations give more. They compare the zero zeta with a Hurwitz zeta function.

Definition 14.1. Define

$$\mathfrak{Z}_N^{\mathrm{abs}}(s) = \sum_{\rho\in\mathcal{Z}(\mathsf{D}_N)}|\rho|^{-s}$$

and

$$\mathfrak{Z}_N^{\mathrm{red}}(s) = \sum_{\lambda\in\Lambda_N}|\lambda|^{-s},$$

with multiplicities. Each term means $e^{-s\log|\rho|}$ or $e^{-s\log|\lambda|}$.

THEOREM 14.2 (Sharp absolute convergence half planes). *The series* $\mathfrak{Z}_N^{\mathrm{abs}}$ *converges absolutely and locally uniformly when*

$$\Re s > 1.$$

It fails to converge absolutely when $\Re s \leq 1$. *The series* $\mathfrak{Z}_N^{\mathrm{red}}$ *converges absolutely and locally uniformly when*

$$\Re s > \frac{1}{N}.$$

It fails to converge absolutely when $\Re s \leq 1/N$. *Both functions are holomorphic on their open half planes of absolute convergence.*

Proof. By theorem 11.14, the zeros of D_N have linear counting density, while the reduced zeros have counting order $R^{1/N}$. Partial summation gives absolute convergence in the stated open half planes. The positive terms obtained after taking absolute values diverge on the boundary and to its left. Local uniform convergence follows by the Weierstrass test on compact subsets. □

THEOREM 14.3 (Meromorphic continuation of the symmetry grouped zeta). *Put*

$$A_N = \frac{\pi}{\sin(\pi/N)}.$$

There is an entire function H_N *such that*

$$\mathfrak{Z}_N^{\mathrm{red}}(s) = A_N^{-Ns}\zeta\left(Ns, \frac{1}{2}\right) + H_N(s)$$

on $\Re s > 1/N$. *The right hand side gives a meromorphic continuation to* $\mathbb{C}$ *with one simple pole at* $s = 1/N$. *Its residue is*

$$\frac{\sin(\pi/N)}{N\pi}.$$

There is an entire function K_N *such that*

$$\mathfrak{Z}_N^{\mathrm{abs}}(s) = NA_N^{-s}\zeta\left(s, \frac{1}{2}\right) + K_N(s).$$

Thus $\mathfrak{Z}_N^{\mathrm{abs}}$ *continues meromorphically to* $\mathbb{C}$ *with one simple pole at* $s = 1$, *of residue*

$$\frac{N\sin(\pi/N)}{\pi}.$$

Proof. By corollary 11.13, all but finitely many reduced zeros can be written

$$\lambda_m = \left(A_N\left(m + \frac{1}{2}\right)\right)^N (1 + \varepsilon_m)$$

with ε_m exponentially small. On every compact set of s values,

$$(1 + \varepsilon_m)^{-s} - 1 = O(|\varepsilon_m|)$$

uniformly for large m. The difference

$$\lambda_m^{-s} - A_N^{-Ns}\left(m+\frac{1}{2}\right)^{-Ns}$$

therefore has an exponentially convergent sum on every compact subset of $\mathbb{C}$. The finitely many exceptional zeros and the finitely many omitted model terms contribute entire functions. This proves the first formula with an entire remainder.

The Hurwitz zeta function has a meromorphic continuation with one simple pole at its first argument equal to one and residue one. Composition with Ns gives the stated pole and residue.

Every nonzero zero λ of Δ_N gives N zeros of D_N, each of modulus $|\lambda|^{1/N}$. Hence

$$\mathfrak{Z}_N^{\mathrm{abs}}(s) = N\mathfrak{Z}_N^{\mathrm{red}}(s/N).$$

Substitution gives the second formula. □

Corollary 14.4 (Exact zeta models in orders two and four). *For $N = 2$,*

$$\mathfrak{Z}_2^{\mathrm{red}}(s) = \pi^{-2s}\zeta\left(2s, \frac{1}{2}\right)$$

and

$$\mathfrak{Z}_2^{\mathrm{abs}}(s) = 2\pi^{-s}\zeta\left(s, \frac{1}{2}\right).$$

For $N = 4$, put $A_4 = \sqrt{2}\pi$. Then

$$\mathfrak{Z}_4^{\mathrm{red}}(s) = A_4^{-4s}\zeta\left(4s, \frac{1}{2}\right)$$

and

$$\mathfrak{Z}_4^{\mathrm{abs}}(s) = 4A_4^{-s}\zeta\left(s, \frac{1}{2}\right).$$

In these two cases the entire correction terms in theorem 14.3 *vanish identically.*

Proof. For $N = 2$, the reduced zeros are

$$\pi^2\left(m+\frac{1}{2}\right)^2, \qquad m \geq 0.$$

For $N = 4$, example 11.12 gives the reduced zeros

$$A_4^4\left(m+\frac{1}{2}\right)^4, \qquad m \geq 0.$$

Summing their negative powers proves the reduced formulas. The absolute formulas follow from

$$\mathfrak{Z}_N^{\mathrm{abs}}(s) = N\mathfrak{Z}_N^{\mathrm{red}}(s/N).$$

□

Definition 14.5. Choose a logarithm for each member of the finite exceptional part of Λ_N. Use the real logarithm for the eventual positive sequence. Define

$$\mathfrak{Z}_N^{\mathrm{orb}}(s) = \sum_{\lambda \in \Lambda_N} e^{-s \operatorname{Log} \lambda}.$$

COROLLARY 14.6 (Branch grouped continuation). *The orbit zeta $\mathfrak{Z}_N^{\mathrm{orb}}$ continues meromorphically to $\mathbb{C}$. It has the same single pole and residue as $\mathfrak{Z}_N^{\mathrm{red}}$. Changing the logarithms of the finite exceptional zeros changes the continuation by an entire function.*

Proof. The tail consists of positive reduced zeros, so it has the same comparison with the Hurwitz zeta. Only finitely many terms depend on the branch choices. □

Definition 14.7. Let q be a positive integer. Choose logarithms of the simple zeros of D_N with arguments in a fixed bounded interval. Define

$$\mathfrak{Z}_{N,q}^{\mathrm{der}}(s) = \sum_{\substack{\mathsf{D}_N(\rho)=0 \\ \mathsf{D}_N'(\rho)\neq 0}} \frac{e^{-s \operatorname{Log} \rho}}{\mathsf{D}_N'(\rho)^q}.$$

THEOREM 14.8 (Derivative weighted convergence). *If $N \geq 3$ and $q \geq 1$, the series $\mathfrak{Z}_{N,q}^{\mathrm{der}}(s)$ converges locally uniformly for every $s \in \mathbb{C}$. It therefore defines an entire function.*

For $N = 2$, the series converges absolutely on $\Re s > 1$.

Proof. For $N \geq 3$, theorem 11.7 gives

$$|\mathsf{D}_N'(\rho)|^{-1} \leq C e^{-c_N |\rho|}$$

for every sufficiently large simple zero. On a compact set of s values, the factor $e^{-s \operatorname{Log} \rho}$ grows at most polynomially in $|\rho|$. Exponential decay therefore gives uniform convergence. The finite exceptional set causes no difficulty.

For $N = 2$, the derivative of the cosine has modulus one at every zero. The assertion then follows from the linear zero spacing. □

Remark 14.9. The derivative weighted sum omits the finitely many zeros that might be multiple. A treatment of those zeros requires the complete principal parts of D_N^{-q} rather than the single derivative weight.

15. **Holomorphic operator substitution**

The scalar identities have a direct functional calculus form. This section records only the basic consequence and does not develop an operator resolvent theory.

Standard Fact 15.1 (Holomorphic functional calculus). Let A be a bounded operator on a complex Banach space. If f is holomorphic on a neighborhood of $\mathrm{Spec}(A)$, then

$$f(A) = \frac{1}{2\pi i} \int_\Gamma f(z)(zI - A)^{-1}\, dz$$

for a contour Γ surrounding the spectrum. The calculus preserves sums, products, and composition. We use the standard form from [DS58].

PROPOSITION 15.2 (Operator reciprocal family). *Let A be bounded and suppose* $\mathrm{Spec}(A)$ *lies in a simply connected region on which* D_N *has no zeros. Fix a holomorphic branch* $g = \log \mathsf{D}_N$ *there. Then*

$$\mathsf{S}_N^{(r)}(A) = \exp\bigl(-r g(A)\bigr) = \bigl(\mathsf{D}_N^{-r}\bigr)(A)$$

is well defined and

$$\mathsf{S}_N^{(r)}(A)\mathsf{S}_N^{(s)}(A) = \mathsf{S}_N^{(r+s)}(A).$$

If (P_i) *is a finite complete family of bounded idempotents, then*

$$P_i f(A) P_j = \frac{1}{2\pi i} \int_\Gamma f(z) P_i (zI - A)^{-1} P_j\, dz$$

for every holomorphic f *on a neighborhood of the spectrum.*

Proof. The spectral mapping theorem makes $\mathsf{D}_N(A)$ invertible. The product identity follows from multiplicativity of the functional calculus and the fixed logarithm. Applying P_i and P_j to the Cauchy integral formula gives the packet formula. □

16. **Further questions**

The large zero theorem settles eventual simplicity, but it does not classify the finite exceptional set. The strongest remaining zero question is whether every zero of D_N is simple and whether every reduced zero of Δ_N is positive. Both statements hold for $N = 2$ and $N = 4$. The argument in this paper proves them outside a compact set for every N.

The first correction in theorem 11.8 can be iterated. Each substitution uses only the finite remainder R_N and its derivatives. This should produce a full exponential expansion for τ_m. Such an expansion would refine the entire correction in the zero zeta formula and could lead to explicit special values.

The symmetry grouped zero zeta has a meromorphic continuation with one pole. A functional equation is not obtained here. The unsymmetrized branch zeta over all N roots of each reduced zero also depends on how logarithms are assigned to the rays. Any proposed functional equation must specify that choice.

The coefficient families in sections 7, 9 and 10 have exact closure laws. Their arithmetic behavior, including odd prime congruences and the separate dyadic weighting, belongs to a different problem. It is not inferred from the complex analytic results proved here.

17. **Conclusion**

Cyclic projection turns one exponential into a finite differential basis. A signed cycle gives the denominator D_N. Reciprocal powers of this denominator form a one parameter semigroup, and their normalized coefficients form a sequence of binomial type. The logarithmic derivative satisfies a higher Riccati equation. Normalized exponential sections produce congruential reciprocal functions and ordinary Appell systems. A scaled reciprocal coupling produces the theta polynomials.

The finite algebra model shows that the differential cycles and addition laws come from one exponential in a quotient algebra. The change of variable $w = z^N$ removes the repeated rotational copies of the zeros. The reduced denominator has order $1/N$, a genus zero Hadamard product, and a trace class determinant model. Its zero power sums are exactly the logarithmic coefficients that govern the transfer tangent and the reciprocal Bell polynomials. Two adjacent exponentials determine every sufficiently large zero. This gives a first displacement term, eventual simplicity, the exact zero counting constant, derivative growth, and meromorphic continuation of the symmetry grouped zero zeta.

The common analytic chain is

$$\begin{aligned}&\text{cyclic projection} \longrightarrow \text{finite differential basis} \longrightarrow \text{reciprocal powers}\\ &\longrightarrow \text{logarithmic coefficients} \longrightarrow \text{zero power sums} \longrightarrow \text{zero zeta.}\end{aligned}$$

Each step is exact under the hypotheses stated in the corresponding theorem.

Use of language model assistance

An OpenAI language model was used as an organizational and mathematical drafting aid. Its contributions included suggestions for the finite algebra formulation in section 6, the reciprocal Bell and Newton formulas in section 7, the Appell operator identities in section 9, the dominant exponential and first displacement arguments in section 11, the trace class model in section 12, and the Hurwitz zeta comparison in section 14. The author assumes full responsibility for checking every statement, proof, and citation in any submitted version. The model is not an author.

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